HyperR3SNet: Leveraging Hyperbolic Space and Vision Foundation Models for Remote Sensing Semantic Segmentation

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***Abstract*—Remote sensing semantic segmentation is driven by land-use monitoring, urban planning, and ecological assessment, yet progress is hampered by scarce pixel-level labels. To address this issue, we present HyperR3SNet, which is an efficient framework for remote sensing semantic segmentation that tackles data scarcity and scale variations in overhead imagery. HyperR3SNet transfers self-supervised vision foundation models (VFMs) to remote sensing, providing strong feature generalization with minimal labeled data. Building on this, a multiscale cross-attention (MSCA) module is inserted into the backbone, Vision Transformer layer, enabling the network to extract richer features across widely varying object scales. To keep the model lightweight, a matrix-factorized task head is employed, sharply reducing parameters and computation while sustaining accuracy. HyperR3SNet is the first model to integrate a hyperbolic pixel-level loss with VFM adaptation for cross-domain and low-annotation remote sensing segmentation, leveraging the exponential expansion property of hyperbolic geometry to capture latent interclass relations and preserve structural consistency under weak supervision. Evaluated on the widely adopted remote sensing segmentation datasets (e.g., iSAID, LoveDA, Potsdam, and Vaihingen), HyperR3SNet achieves mean intersection over union (mIoU) of 67.60%, 55.86%, 80.07%, and 96.46%, respectively, and on average surpasses a broad range of state-of-the-art (SOTA) methods.**

***Index Terms*—Domain adaptation, hyperbolic embedding, low-rank decomposition, remote sensing segmentation, vision foundation models (VFMs).**

I. Introduction

REMOTE semantic (RS) segmentation is fundamental to remote sensing applications, including land-cover mapping, urban planning, and disaster response. Recent advances in self-supervised vision foundation models (VFMs), trained on large-scale natural-image corpora, have yielded highly transferable features for segmentation tasks [1], [2], [4], [5], [7]. However, progress in remote sensing remains limited by the scarcity of pixel-level annotations and the substantial domain gap between aerial imagery and natural photographs. Differences in semantic composition, object scale, and spectral characteristics lead to significant performance degradation when natural-image pretrained weights are directly applied. In addition, remote sensing datasets and domain-specific VFMs are relatively small compared with their natural-image counterparts [9], [10], [11], [12], a limitation further reinforced by confidentiality constraints in military and commercial contexts. Although unsupervised VFMs pretrained on natural images demonstrate promising transferability to remote sensing [13], the development of efficient adaptation strategies to fully exploit their potential has become a pressing research priority.

Adapting VFMs from the natural-image domain to remote sensing scenarios still confronts three prominent challenges:

1) High-quality pixel-level annotations in remote sensing are extremely scarce—typically only at the scale of hundreds of thousands, which is three to four orders of magnitude fewer than in natural-image datasets. Such scarcity makes direct training from scratch infeasible, and full fine-tuning of large models initialized with natural-image weights is both inefficient and computationally costly.
2) The domain gap induced by differences in spectral bands, resolution, and viewpoint distortions causes conventional Euclidean feature spaces to fail in capturing transferable representations, thereby limiting expressive power.
3) Remote sensing categories often exhibit implicit tree-structured hierarchies, where semantically related classes (e.g., "Forest" and "Shrub") share latent parent concepts (e.g., "Vegetation"), while "Building" and "Road" share the parent concept "Man-made Structures."

Received 21 November 2025; revised 31 March 2026; accepted 18 April 2026. Date of publication 23 April 2026; date of current version 7 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42330707, in part by the Natural Science Foundation of Zhejiang Province under Grant LZ25F010007, and in part by the Artificial Intelligence Special Project under Grant ky202525. *(Junjie Fu and Chenliang Wang are co-first authors.) (Corresponding author: Chenliang Wang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3687072

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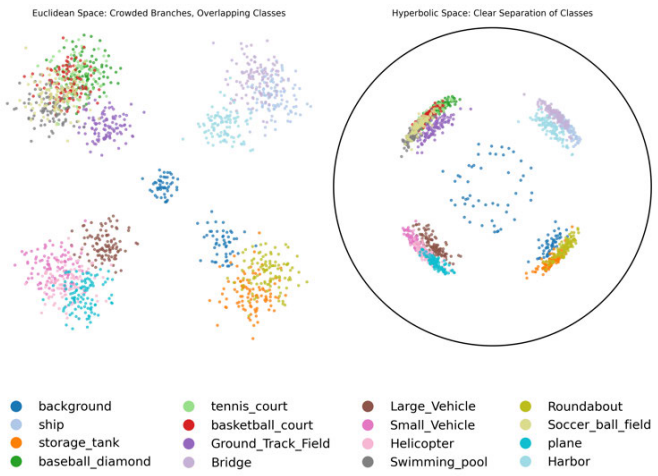


Fig. 1. Comparison of class distributions in Euclidean versus hyperbolic embedding spaces.

Yet, Euclidean embeddings flatten these relationships because Euclidean geometry expands polynomially, which is insufficient to represent the exponential growth of tree-like hierarchies without severe distortion. This flattening exacerbates parent-class bias and long-tail misclassification, while pronounced scale variation and geometric distortions further undermine the single-scale or fixed-pattern attention of vanilla vision transformer (ViT) architectures. Taken together, these factors form a *coupled triad of errors*: *statistical error* from scarce annotations, *representation error* amplified by domain shifts, and *structural error* introduced by flattening hierarchical semantics. Conventional remedies attempt to offset these errors with larger models and more data, but such strategies often lead to a "compute-for-accuracy" dead end.

Taken together, these issues can be viewed as a *coupled triad of errors*: *statistical error* caused by scarce annotations, *representation error* induced by domain gaps when transferring natural-image weights, and *structural error* from flattening hierarchical categories in an Euclidean space. This triad of errors is mutually reinforcing, making remote sensing segmentation particularly challenging under weak supervision. It is important to clarify that throughout this article, we use the term "weak supervision" to denote the regime of label scarcity (i.e., insufficient quantity of pixel-level annotations), distinguishing it from the traditional definition involving incomplete supervision signals (e.g., image-level tags or scribbles).

Hyperbolic geometry offers a compelling alternative. Its negative curvature provides exponential capacity to embed hierarchical structures in low-dimensional space. The hyperbolic distance amplifies semantic gaps between classes, while gyroplane decision boundaries naturally align with hierarchical semantics. As illustrated in Fig. 1, this visualization demonstrates why hyperbolic embeddings are particularly suited for hierarchical data such as remote sensing object categories with natural taxonomic relationships. The Poincaré disk model visually represents infinite hyperbolic space within a bounded disk, where the distortion increases toward the boundary to accommodate the exponential growth of available space. Through these properties, the dependence on large-scale pixel annotations can be reduced, and strong discrimination with structural consistency can still be achieved under weak supervision. The strengths of Euclidean pretrained backbones are not discarded. Natural-image VFMs (e.g., DINOv2 [7] and DINOv3 [8]) already provide robust and transferable representations. Direct projection into hyperbolic space at the feature-extraction stage would introduce unnecessary complexity and may harm generalization.

To overcome these limitations mentioned in the preceding text, we exploit geometric priors rather than parameter redundancy—shifting the focus from *how much computation* to *which space to learn in*. This principle is operationalized through a three-stage collaborative framework. A *frozen encoder* retains universal Euclidean representations to mitigate statistical error without retraining. A *multiscale cross-attention (MSCA)* adapter corrects scale variation and geometric deformation to alleviate representation error via cross-domain alignment. Finally, a *hyperbolic projection* introduces structural priors through exponential mapping and gyroplane classification, thereby resolving structural error and preserving hierarchical semantics. Together, these stages share a unified objective and form an end-to-end geometry-parameter synergy, enabling efficient training with only 16M parameters while surpassing much larger baselines.

In summary, our technical contributions include the following.

1) We propose HyperR3SNet, a new and efficient framework for remote sensing semantic segmentation that, for the first time in the field of remote sensing, combines a VFM with hyperbolic geometry, enabling precise modeling of complex scene semantics under weak supervision.
2) The framework incorporates four key innovations: a) effective feature transfer that leverages large-scale self-supervised weights from natural-image models to improve generalization under limited labels; b) an MSCA layer that captures fine spatial details across diverse object scales and shapes; c) a matrix-factorized lightweight head that sharply reduces parameters and computation for greater efficiency; and d) a pixel-level hyperbolic feature loss that exploits latent interclass structure, strengthening semantic discrimination and structural awareness even with flat taxonomies and scarce supervision.
3) Extensive experiments on iSAID, LoveDA, Potsdam, and Vaihingen compare HyperR3SNet with nine public state-of-the-art (SOTA) segmentation methods. On the mean intersection over union (mIoU) metric, the average improvement was 9.62%, 3.94%, 2.24%, and 5.36%, respectively, demonstrating superior performance and generalization ability.

The remaining sections of this article are outlined as follows. Section II reviews related work on VFMs, semantic-segmentation methods, multiscale attention mechanisms, and hyperbolic embeddings. Section III details the proposed HyperR3SNet framework, including its MSCA module, lightweight head design, and hyperbolic feature projection. Section IV presents the experimental results and ablation

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studies on multiple remote sensing datasets. Finally, Section V concludes this article.

II. Related Work

In this section, we organize related work into four parts:

1) *VFMs*, where we review self-supervised and vision–language pretraining in both natural and remote sensing imagery to clarify the backbone choices for our framework;
2) *semantic-segmentation methods*, including convolutional neural network (CNN)- and Transformer-based architectures, with a focus on their strengths and limitations when transferred to remote sensing scenarios;
3) *multiscale and deformable-attention mechanisms*, which are crucial for handling severe scale variation and geometric distortions in aerial and satellite images;
4) *hyperbolic space embedding*, where we summarize recent progress in hyperbolic representation learning for computer vision and remote sensing, and position our pixel-level hyperbolic modeling relative to these works.

A. Vision Foundation Model

In recent years, self-supervised learning has made remarkable strides in the natural-image domain, creating valuable opportunities for transfer to remote sensing semantic segmentation. MoCo [14] employs a dynamic dictionary for contrastive learning, achieving efficient and stable representation training with modest memory usage; SimCLR [15] attains high-quality features through extensive data augmentation and a contrastive loss; masked autoencoder (MAE) [3] encourages models to learn rich visual semantics by randomly masking and reconstructing image pixels; EVA02 [1] further enhances representational capacity and efficiency with an improved Vision Transformer architecture; and The Dino Series [6], [7], [8] pushes the generalization of self-supervised features to new heights by coupling an optimized self-distillation strategy with a refined training pipeline. Meanwhile, the cross-modal model CLIP [4] greatly boosts the transferability of visual representations, and SAM [5] demonstrates outstanding zero-shot segmentation and generalization. Together, these SOTA VFMs open multiple efficient transfer pathways for remote sensing semantic segmentation.

Meanwhile, several foundation models tailored to satellite imagery have emerged within the remote sensing community. SeCo [16] is the first to adapt the MoCo contrastive-learning framework to remote sensing, pairing it with augmentations designed for aerial data to achieve effective self-supervised representation learning. SatMAE [17] extends the MAE paradigm to satellite images and obtains high-quality features by reconstructing hidden regions. RingMo [10] employs a cross-modal masked-autoencoding strategy that integrates optical and synthetic-aperture radar (SAR) signals, demonstrating strong, general-purpose representations. RemoteCLIP [9] introduces vision–language contrastive learning to remote sensing tasks and delivers excellent performance in semantic understanding and image retrieval, while Falcon [18] explores deeper visual–text fusion, providing a new avenue for satellite-image interpretation. Although these domain-specific models have advanced remote sensing semantic segmentation, their representational power is still limited by the modest scale and diversity of available satellite data.

B. Semantic-Segmentation Method

In remote sensing image interpretation, semantic-segmentation tasks still commonly rely on CNN architectures. The DeepLab family [21], [22], [23], [24] captures multiscale context with Atrous convolutions and an Atrous spatial pyramid pooling (ASPP) module; HRNet [60] preserves fine spatial details through high-resolution representations and parallel multiresolution branches; UNet [28], featuring an encoder–decoder structure, is widely used in medical and remote sensing tasks for its effective fusion of shallow and deep features; PSPNet [29] introduces a pyramid pooling module to gather global context and improve holistic scene understanding; and SegNet [30] restores fine details efficiently by retaining max-pooling indices during upsampling. Although each of these CNN models offers specific advantages, they rely on fixed kernels and static receptive fields, making them less adaptable to the extreme scale variation and geometric distortions inherent in remote sensing imagery, and thus limiting their generalization capacity.

Transformer-based architectures have recently demonstrated impressive capabilities in computer vision. Vision Transformer [31] captures long-range dependencies by applying self-attention to image patches, thereby enhancing global semantic awareness. Swin Transformer [32] introduces a hierarchical design with shifted windows to accommodate multiscale features. Building on these advances, Segmenter [33] and SETR [34] pioneer the use of Transformers in semantic segmentation, employing an encoder–decoder layout with self-attention to fuse global context, while SegFormer [35] further reduces parameters and computation through a lightweight Transformer design. Nevertheless, these models still incur high computational cost and rely on fixed attention windows, which limit their generalization when confronted with the pronounced scale variation and geometric distortions of remote sensing imagery.

C. Multiscale and Deformable-Attention Mechanism

Multiscale feature-fusion approaches—exemplified by PSPNet [29] and DeepLabv3+ [24]—significantly boost segmentation accuracy, especially on remote sensing images that exhibit pronounced scale variation. In parallel, methods such as deformable DETR [19] and the deformable-attention Transformer [38] introduce adaptive sampling, offering greater spatial flexibility that is inherently well-suited to remote sensing scenarios. Nevertheless, the use of deformable attention for remote sensing semantic segmentation remains largely unexplored.

D. Hyperbolic Space Embedding

Because of its negative curvature, hyperbolic space embeds tree-like and hierarchical structures with exponential efficiency

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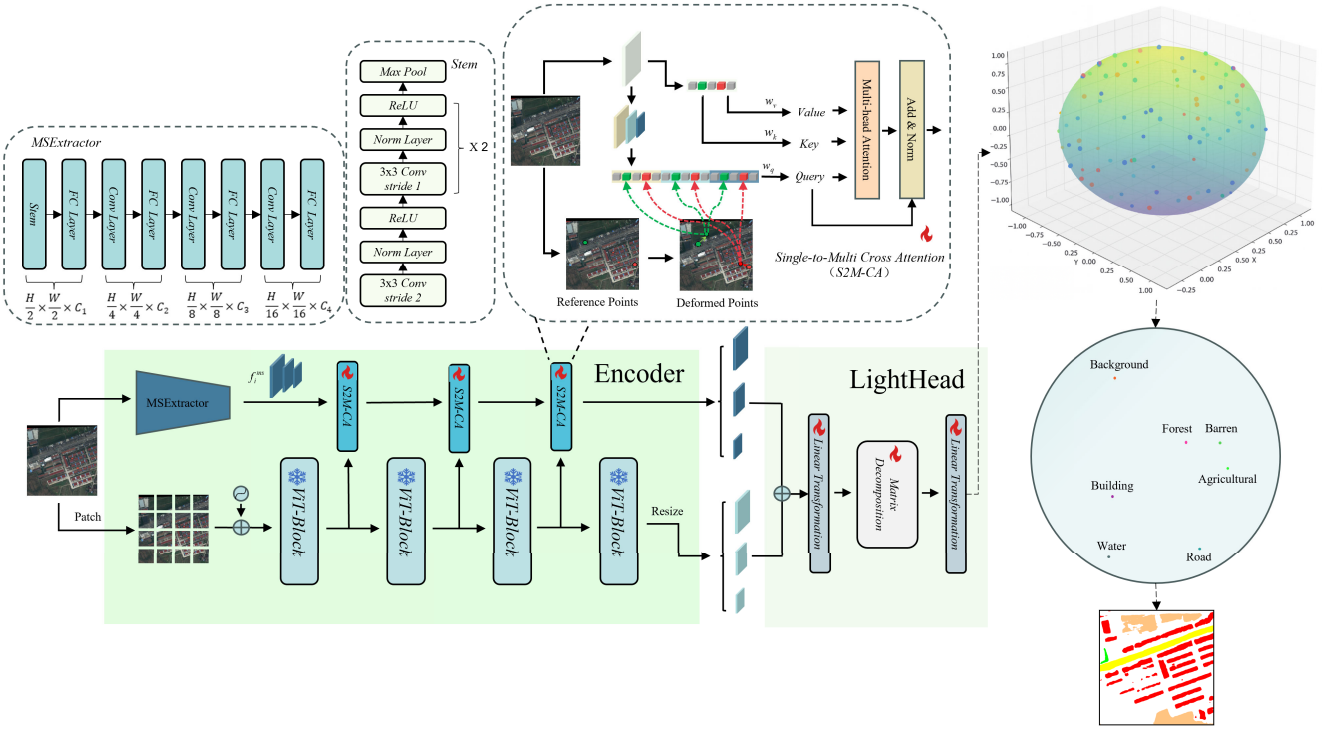


Fig. 2. Overall architecture of the proposed HyperR3SNet framework. Modules marked with the flame icon indicate trainable components, while modules marked with the snowflake icon denote frozen components during training.

and has been widely adopted in natural-language processing (e.g., Poincaré embeddings [39]) and knowledge-graph modeling (e.g., HyperKG [40]). Computer-vision research is now following suit: Khrulkov et al. [48] pioneered hyperbolic image embeddings, and subsequent studies have explored hyperbolic geometry for image classification [41], [42], image retrieval [43], [45], and, albeit still nascent, semantic segmentation [44], [46], [47]. In remote sensing, Cui et al. [51] proposed a hyperbolic embedding model for scene classification, while Li et al. [56] introduced the first fully hyperbolic neural network for remote sensing semantic segmentation, which augments hyperbolic representations with Euclidean affinity constraints to improve optimization stability and discriminability. In the context of point cloud matching, a hyperbolic embedding and confident patch-guided network [49] further leverages hyperbolic attention to exploit the inherent hierarchical structure of terrestrial laser scanner point clouds. Despite these advances, systematic explorations of hyperbolic geometry at the pixel level remain limited. To bridge this gap, we introduce hyperbolic embeddings into remote sensing semantic segmentation for the first time and devise a pixel-level hyperbolic loss that explicitly captures latent semantic hierarchies, thereby enhancing structural awareness and generalization under weak supervision and flat label taxonomies.

III. Method

To tackle the challenges of sparse supervision, large intra-class variance, and weak semantic structure in remote sensing semantic segmentation, we propose HyperR3SNet, an efficient framework that combines the representational power of VFMs with the hierarchical modeling capacity of hyperbolic space. This section proceeds as follows. We begin with an overview of the HyperR3SNet architecture, clarifying its guiding rationale and the interplay among its principal components. Next, we present the MSCA module, explaining how it captures fine-grained contextual information across multiple spatial scales in remote sensing imagery. Finally, we describe the lightweight decoder head and the associated pixel-level hyperbolic loss, illustrating how feature learning in hyperbolic space reinforces semantic hierarchy awareness and improves generalization.

A. Overall Architecture

In this section, we introduce an efficient semantic-segmentation network built on the powerful feature-extraction VFM. As illustrated in Fig. 2, the framework contains four key components: 1) a feature encoder derived from a VFM; 2) a multiscale deformable-attention module; 3) a lightweight task head; and 4) a pixel-level hyperbolic supervision branch. These modules work in concert to strengthen feature representation, enable precise modeling of semantic structure, and preserve high inference efficiency.

A core design philosophy of HyperR3SNet is the intentional separation of feature extraction (Euclidean) and semantic classification (hyperbolic). We retain the encoder in Euclidean space because remote sensing imagery resides on a regular grid, where Euclidean operations (e.g., convolutions and patch embeddings) are highly optimized for extracting local visual patterns like textures and edges. However, as features

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propagate to the final stage, they represent abstract semantic concepts with latent hierarchical structures (e.g., "Water" versus "River"). Mapping these high-level features to the Poincaré Ball allows the classifier to exploit the exponential volume of hyperbolic space, effectively modeling these semantic hierarchies and uncertainties that Euclidean spaces struggle to represent.

Motivated by the goal of achieving superior generalization with stronger pretraining and fewer trainable parameters, we adopt the cutting-edge VFM DINOv3 as a *frozen* feature encoder for remotesensing semantic segmentation. Compared with conventional CNNs (e.g., ResNet [50]) and early Transformers (e.g., ViT [31]) that are trained under full supervision, self-supervised foundation models learn richer, more transferable representations from large natural-image corpora, making them particularly effective for small-sample or weakly supervised tasks [6], [7], [8]. Thanks to an enhanced self-distillation framework and a data-intensive training regimen, DINOv3 delivers representation power and generalization that surpass its predecessors. Nevertheless, its ViT-based single-scale attention is suboptimal for the pronounced scale variation in remote sensing imagery. To address this limitation, we embed an MSCA layer between backbone blocks, injecting multiscale information while adding only a small number of additional parameters and thereby improving feature adaptability. In the decoder, we model global context as a low-rank recovery problem and employ a matrix-factorization head to learn features in a low-rank subspace, which is both efficient and highly parameter-economic. Finally, to exploit the inherent semantic hierarchy of remote sensing scenes, we project these low-rank features into hyperbolic space and perform hierarchical segmentation on the Poincaré Ball, further boosting accuracy and generalization.

B. Multiscale Cross-Attention

In order to generate multiscale information from the input features of the image, we used standard convolution layers from ResNet [50] to build a simple multiscale feature extractor. Given an input image $\mathbf{X}_{\text{rgb}} \in \mathbb{R}^{B\times 3\times H\times W}$, we first perform shallow feature extraction on the image using a multiscale feature-extractor module (MSExtractor)

$$\left\{F_1^i\right\} = F_{ms}\left(X_{rgb}\right) \tag{1}$$

where F_{ms} denotes $\Phi_{\text{MSExtractor}}$. This module receives the input remote sensing image $\mathbf{X}_{\text{rgb}}$ and extracts a set of feature tensors with different semantic depths and spatial resolutions through layer-by-layer downsampling and channel expansion: $\{F_1^i \in \mathbb{R}^{H/(2^{i+1})\ \times\ W/(2^{i+1})\ \times\ d_1}\}$, $i \in \{1, 2, 3, 4\}$.

Formally, a standard ViT partitions the input image $x \in \mathbb{R}^{H\times W\times C}$ into patches of size $P \times P$, resulting in $N = HW/P^2$ patches, which are then embedded as

$$\mathbf{z}^{(0)} = \left[x^1\mathbf{E};\ x^2\mathbf{E};\ \ldots;\ x^N\mathbf{E}\right] + \mathbf{E}_{\text{pos}} \tag{2}$$

where x^i is the ith patch, $\mathbf{E}$ is the linear embedding, and $\mathbf{E}_{\text{pos}}$ denotes the positional encoding.

Standard self-attention is computed as

$$\text{Attention}\left(Q, K, V\right) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \tag{3}$$

where Q, K, and V are the query, key, and value matrices, respectively, and d_k is the key dimension.

To address the issue of the single feature size used in visual foundation models, inspired by Deformable DETR [19], a single-to-multi-cross-attention (S2MCA) layer was designed, introducing multiscale information into the features through a deformable-attention mechanism. To further efficiently learn multiscale features, first, we concatenate the features of different sizes extracted from the MSExtractor into a sequence to construct a global multiscale feature as the value. Second, we use the feature sequence of a specific size output by the original ViT block as the query, introduce the deformable cross-attention mechanism, and calculate the attention to fuse the global features of multiple scales.

Third, cross-attention calculations are performed between the raw ViT module features extracted at each stage and the multiscale features. This ultimately fuses the multiscale feature information with the global feature information from the original ViT output. To fuse spatial features from different scales, we flatten each scale feature map $\mathbf{F}_i \in \mathbb{R}^{B\times C\times H_i\times W_i}$ into a sequence format. The specific calculation steps for the deformable cross-attention mechanism are as follows:

$$\mathbf{F}_i' = \text{Flatten}\left(\mathbf{F}_i\right) \in \mathbb{R}^{B\times(H_iW_i)\times C}, \quad i = 1, \ldots L \tag{4}$$

where L denotes the number of layers in the multiscale feature map. Subsequently, we concatenate all scales of features after aligning them in the channel dimension to obtain a unified input sequence

$$\mathbf{X} = \text{Concat}\left(\mathbf{F}_1', , \mathbf{F}_2', \ldots, \mathbf{F}_L'\right) \in \mathbb{R}^{B\times L'\times C}. \tag{5}$$

The total length of the spliced sequence is

$$L' = \sum_{i=1}^{L} H_iW_i. \tag{6}$$

Given a query tensor $\mathbf{q} \in \mathbb{R}^{N\times L\times C}$ and a set of reference points $\mathbf{r} \in \mathbb{R}^{N\times L\times 2}$, the deformable cross-attention module computes sampling offsets $\Delta\mathbf{p}$ and attention weights $\mathbf{W}$ from the query via two linear projections

$$\Delta\mathbf{p} = \text{Linear}_{\text{offset}}\left(\mathbf{q}\right) \in \mathbb{R}^{N\times L_q\times 2} \tag{7}$$

$$\mathbf{W} = \text{Softmax}\left(\text{Linear}_{\text{attn}}\left(\mathbf{q}\right)\right) \in \mathbb{R}^{N\times L_q} \tag{8}$$

where N is the batch size, L_q is the number of queries, L is the number of feature levels, H is the number of attention heads, and K is the number of sampling points per head.

Sampling locations are then computed by

$$\mathbf{p}_{\text{sample}} = \mathbf{p} + \frac{\Delta p}{\text{Normalizer}\left(H_l, W_l\right)} \tag{9}$$

where spatial shapes (H_l, W_l) of each level are used to normalize the offsets

$$\begin{aligned}&\text{S2MCAttn}\left(\mathbf{Q_s}, \mathbf{K_m}, \mathbf{V_m}, \Delta\mathbf{p}\right)\\ &= \sum_{k=1}^{K} \text{Softmax}\left(\frac{\mathbf{Q}\cdot\mathbf{K}_{\mathbf{p}_{\text{sample}}}{}^\top}{\sqrt{d_k}}\right)\cdot\mathbf{V}_{\mathbf{p}_{\text{sample}}}.\end{aligned} \tag{10}$$

This attention structure enables the model to selectively focus on a small number of key positions across feature pyramid levels, significantly reducing computation while

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enhancing geometric adaptability and multiscale context reasoning.

This design enables the model to retain the robust global context modeling of the pretrained ViT while gaining the flexibility to adapt to scale variation and spatial deformation inherent in remote sensing imagery.

C. Light Hyperhead

In the head section, we aim to design a lightweight decoder. Inspired by Geng et al.'s [55] work, we realize that it is unnecessary to use multilayer attention mechanisms for decoding. The essence of the Transformer is to model the dependencies between tokens through attention, and matrix decomposition can also be used in the same way. This design is more lightweight and efficient. Therefore, this article proposes a lightweight segmentation head based on matrix decomposition and hyperbolic geometry. We first use a lightweight matrix decomposition method to efficiently compress the features extracted by the encoder, reducing redundant calculations and focusing on the main feature information. Then, we introduce a hyperbolic geometry projection operation to map low-rank semantic features to a negative curvature space, thereby more effectively modeling the hierarchical structure and nested relationships between remote sensing semantic categories.

1) Matrix Decomposition: In order to model the global contextual information in images more effectively, we analyze the structural properties of the feature space. Let the input image be encoded by the encoder, and the output feature map tensor be $\mathcal{X} \in \mathbb{R}^{C\times H\times W}$, where C denotes the number of channels, and $H\times W$ denotes the spatial dimensions. We unfold it into a matrix $\mathbf{X} \in \mathbb{R}^{C\times HW}$, where each column represents the superpixel features of a spatial location.

The theoretical rationale for treating $\mathbf{X}$ as a low-rank matrix stems from the inherent spatial redundancy in remote sensing imagery. Unlike natural images with high-frequency clutter, RS scenes are typically dominated by large, continuous homogeneous regions (e.g., vast agricultural fields, water bodies, or forests). Pixels within these regions share highly correlated spectral and semantic patterns, resulting in significant linear dependence among their feature vectors. Furthermore, since the number of semantic categories is vastly smaller than the number of pixels ($K \ll HW$), the high-dimensional feature space is effectively spanned by a small set of basis vectors representing these semantic prototypes. This implies that $\mathbf{X}$ resides on a low-dimensional manifold, such that there exists a basis matrix $\mathbf{D} \in \mathbb{R}^{C\times r}$ and a coefficient matrix $\mathbf{C} \in \mathbb{R}^{r\times HW}$ satisfying

$$\mathbf{X} \approx \mathbf{DC} \tag{11}$$

where $r \ll HW$, i.e., $\mathbf{X}$ is low rank. This modeling approach not only extracts global semantic components but also removes redundancy and noise. In this article, we first perform a linear mapping of the feature dimensions to reduce the number of hidden layer features, and then use matrix decomposition to learn low-rank space features. Finally, we apply a linear layer to perform a linear transformation on the low-rank signal

$$\hat{\mathbf{Z}} = W_u \cdot \mathcal{M}(W_l \cdot \mathbf{Z}) \tag{12}$$

where $\mathbf{Z} \in \mathbb{R}^{C\times HW}$ denotes the matrix obtained by flattening the input feature map, where C is the number of channels and HW represents the flattened result of the spatial dimension. $W_l \in \mathbb{R}^{d\times C}$ is a linear transformation matrix used to reduce the input features from C channels to a d-dimensional latent space. $\mathcal{M}(\cdot)$ denotes the matrix decomposition operation module, where nonnegative matrix decomposition is used in this article, with the input being the reduced-dimensional features and the output being a low-rank representation. $W_u \in \mathbb{R}^{C\times d}$ is an upsampling matrix used to remap the low-rank features back to the original number of channels. The final output is $\hat{\mathbf{Z}} \in \mathbb{R}^{C\times HW}$, whose structure remains consistent with the input.

D. Hyperbolic Space Feature

Traditional hierarchical segmentation methods attempt to explicitly model this parent–child class structure in order to enhance semantic consistency and generalization capabilities. However, such methods often face the problem of parent bias [47] in cross-domain transfer learning scenarios. This problem is caused by the uneven distribution of decision boundaries between different classes in the feature space, which makes parent class predictions prone to bias toward a certain child class cluster, leading to a decline in structural generalization capabilities.

Hyperbolic space is a non-Euclidean space with constant negative curvature, whose most distinctive feature is that its spatial capacity grows exponentially with radius. This characteristic is crucial for alleviating the semantic "flattening" distortion inherent in Euclidean embeddings, which manifests most severely in three practical scenarios within remote sensing imagery. First, it addresses *hierarchical ambiguity*, where parent categories (e.g., "Vegetation") and their child nodes (e.g., "Forest" versus "Shrub") are often compressed into equidistant clusters in Euclidean space, causing the model to lose the inclusion relationship. Second, it mitigates the impact of *long-tail distributions*, preventing high-frequency classes (e.g., "Background") from dominating the feature volume and squeezing minority classes (e.g., "Cars") into narrow margins. Finally, for *visually similar class boundaries*—such as "Road" versus "Roof" within impervious surfaces—hyperbolic geometry provides sufficient margin separation that Euclidean metrics often fail to achieve. Consequently, by projecting features into the Poincaré Ball, we leverage the negative curvature to allocate exponentially more volume for these hierarchical and boundary-sensitive cases.

Among the many hyperbolic geometric models, we chose to model on the *Poincaré Ball model*. The *Poincaré Ball* is defined as an n-dimensional Riemannian manifold $(\mathbb{D}_c^n, g_{\mathbb{D}_c^n})$ with curvature $-c$, where

$$\mathbb{D}_c^n = \left\{ x \in \mathbb{R}^n \,\middle|\, c\|x\|^2 < 1 \right\}. \tag{13}$$

This measure ensures that the distance increases rapidly when approaching the ball boundary, thereby naturally enhancing the distinction between classes.

In order to map Euclidean vectors to hyperbolic space, we introduce the exponential map, which is a key operation for implementing Euclidean feature embedding into hyperbolic

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space. For any point $x \in \mathbb{D}_c^n$, the exponential map $\exp_x^c : T_x\mathbb{D}_c^n \longrightarrow \mathbb{D}_c^n$, its exponential map form under the Poincaré Ball model is defined as follows:

$$\exp_x^c(v) = x \oplus_c \left(\tanh\left(\frac{\sqrt{c}\lambda_x^c \|v\|}{2} \right) \frac{v}{\sqrt{c}\|v\|} \right) \tag{14}$$

where $\lambda_x^c = 2/(1 - c\|x\|^2)$ is the scaling factor at the anchor point x, and v is the Euclidean vector in the tangent space at that point. For a pair $x, y \in \mathbb{D}_c^n$, the symbol $\oplus_c$ denotes the Möbius addition operation in hyperbolic space, which is defined as

$$x \oplus_c y = \frac{\left(1 + 2c\langle x, y\rangle + c\|y\|^2\right) x + \left(1 - c\|x\|^2\right) y}{1 + 2c\langle x, y\rangle + c^2\|x\|^2\|y\|^2}. \tag{15}$$

This operation defines the translation mechanism in hyperbolic space, ensuring that the result remains within the unit sphere $\mathbb{D}_c^n$.

When the anchor point is the origin ($x = 0$), namely, for $v \in T_x\mathbb{D}_c^n$, the exponential map has a more concise closed-form expression

$$\mathrm{Exp}_0^c(v) = \tanh\left(\sqrt{c}\|v\|\right) \cdot \frac{v}{\sqrt{c}\|v\|}. \tag{16}$$

This mapping ensures that $\mathrm{Exp}_0^c(v) \in \mathbb{D}_c^n$, which can be used as an embedding function from Euclidean space to hyperbolic space to achieve end-to-end optimization during training. This article is also based on this mapping conversion from Euclidean space to hyperbolic space.

In hyperbolic space, we use hyperbolic space to measure the distance between two points. In the Poincaré Ball model, given two points $x, y \in \mathbb{D}_c^n$ located in hyperbolic space $\mathbb{D}_c^n$, the hyperbolic distance between them is defined as follows:

$$d_{\mathbb{H}}(x, y) = \operatorname{arcosh}\left(1 + \frac{2\|x - y\|^2}{\left(1 - \|x\|^2\right)\left(1 - \|y\|^2\right)}\right) \tag{17}$$

where x and z denote the two points in an n-dimensional hyperbolic space with curvature $-c$, such that $\|x\| < 1/\sqrt{c}$ and $\|z\| < 1/\sqrt{c}$. The symbol $\|\cdot\|$ denotes the Euclidean norm, i.e., the length of a vector, so $\|x - z\|^2$ denotes the Euclidean square distance between the two points. The terms $(1 - \|x\|^2)$ and $(1 - \|z\|^2)$ in the denominator are scaling factors for each point's distance from the boundary of the sphere, reflecting the degree of spatial curvature in the Poincaré Ball model: as points approach the boundary, this factor decreases, causing the hyperbolic distance to increase rapidly. The function $\operatorname{arcosh}(\cdot)$ is the inverse hyperbolic cosine function, used to convert hyperbolic cosine values into actual distance values. Finally, the constant $c > 0$ represents the absolute value of negative curvature, controlling the overall expansion rate of space; the larger the c is, the stronger the spatial curvature and the faster the distance growth. The geometric significance of this distance function is as follows: when two points are close to the origin, $d_{\mathbb{H}}(x, z)$ is very close to the Euclidean distance; however, when any point is close to the boundary, the hyperbolic distance grows exponentially, reflecting the property of "infinite distance to the boundary" in hyperbolic space. This characteristic is particularly suitable for modeling data with hierarchical structures or exponential expansion features.

In Euclidean space, classification boundaries are typically represented by hyperplanes $\langle x, a\rangle + b = 0$. However, in hyperbolic space with negative curvature, lines and planes no longer hold, so it is necessary to introduce the corresponding nonlinear structure *Gyroplane*. For each category k, we define the corresponding gyroplane as

$$\mathcal{H}_k^c = \left\{ h \in \mathbb{D}_c^n \;\middle|\; \langle -r_k \oplus_c h,\ a_k\rangle = 0 \right\}. \tag{18}$$

Based on the above definition of gyroplane, we can further calculate the hyperbolic distance from any point $h \in \mathbb{D}_c^n$ to the gyroplane $\mathcal{H}_k^c$ of category k. This distance reflects the geometric relationship between the current point and the boundary of the category in hyperbolic space, and is defined as follows:

$$d_{\mathbb{H}}\left(h, \mathcal{H}_k^c\right) = \frac{1}{\sqrt{c}} \operatorname{arcsinh}\left(\frac{2\sqrt{c}\,|\langle -r_k \oplus_c h, a_k\rangle|}{\left(1 - c\|-r_k \oplus_c h\|^2\right)\|a_k\|} \right). \tag{19}$$

Here, $\langle \cdot, \cdot \rangle$ is the Euclidean inner product. The term $1 - c\|-r_k \oplus_c h\|^2$ in the denominator is the compression factor of the spherical boundary, reflecting the deformation rate of the boundary in hyperbolic geometry. $\operatorname{arcsinh}(\cdot)$ is the inverse hyperbolic sine function, used to convert the gyroplane inner product distance back to the actual geometric distance.

This distance function characterizes the separation between point h and the class boundary, and can be further used to construct a distance-based logit function. In hyperbolic multiclass classification, we convert this distance to the logit score of class k, defined as

$$\zeta_k(h) := \frac{\lambda_{r_k}^c \|a_k\|}{\sqrt{c}} \cdot \operatorname{arcsinh}\left(\frac{2\sqrt{c}\,\langle -r_k \oplus_c h, a_k\rangle}{\left(1 - c\|-r_k \oplus_c h\|^2\right)\|a_k\|} \right) \tag{20}$$

where $\lambda_{r_k}^c = 2/(1 - c\|r_k\|^2)$ is the hyperbolic scaling factor at point r_k, which is used to adjust the local geometric scale at different positions.

Finally, we input this logit into the softmax function to get the category probability

$$p(\hat{y} = k \mid h) \propto \exp\left(\zeta_k(h)\right). \tag{21}$$

Algorithm 1 Hyperbolic Learning Pipeline

Require: Euclidean features $\mathbf{Z} \in \mathbb{R}^{C \times HW}$ (from Encoder).
Require: Curvature c, Gyroplane parameters $\{a_k, r_k\}_{k=1}^K$.
Ensure: Pixel-wise Class Probabilities $\mathbf{P}$.

Step 1: Low-Rank Feature Refinement (Sec. III-C)
  Apply matrix decomposition to remove redundancy:
  $\hat{\mathbf{Z}} \leftarrow W_u \cdot \mathcal{M}(W_l \cdot \mathbf{Z})$ {Eq. 12}
Step 2: Euclidean-to-Hyperbolic Projection (Sec. III-D)
  Map refined features to the Poincaré ball $\mathbb{D}_c^n$:
  $\mathbf{h} \leftarrow \exp_0^c(\hat{\mathbf{Z}})$ {Eq. 16}
Step 3: Gyroplane-based Classification
  Compute distance-based logits for each class k:
  for $k = 1$ to K **do**
    $\zeta_k(\mathbf{h}) \leftarrow \cdots$ {Eq. 20}
  end for
Step 4: Probability Estimation
  $\mathbf{P} \leftarrow \mathrm{Softmax}(\{\zeta_1, \ldots, \zeta_K\})$
return P

To provide a clearer overview of the entire inference process, we summarize the transition from Euclidean feature

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extraction to hyperbolic classification in Algorithm 1. This workflow illustrates how the low-rank refined features are projected onto the Poincaré Ball and subsequently classified using gyroplane boundaries.

The above formulation forms the core of *hyperbolic multinomial logistic regression (H-MLR)*, which achieves a differentiable multiclass classifier design while maintaining the hyperbolic geometry constraints, and is particularly suitable for dealing with semantic labels with hierarchical structure or category spaces with nested relationships.

To alleviate the parent-class bias problem [44], [46], [47], we adopt a hierarchical modeling strategy in the Poincaré Ball. By exploiting the boundary contraction and distance-equilibrium properties of hyperbolic space, semantic categories are embedded more uniformly in low-dimensional space, substantially reducing parent-class prediction errors. It is important to clarify the notion of "flattening" in this context. In the Introduction, flattening referred to the limitations of Euclidean embeddings, which force hierarchical semantics into a flat space and thereby induce structural error. In contrast, remote sensing datasets themselves are annotated with a flat taxonomy, without explicit parent–child supervision signals. Rather than constructing an artificial label hierarchy, we retain this flat annotation setting and rely on hyperbolic geometry to recover latent hierarchical structure implicitly. Specifically, subclass predictions are made in the Poincaré Ball through a gyroplane classifier, whose decision boundaries naturally align with semantic hierarchies. This design yields a compact yet hierarchy-aware framework that improves parent-class prediction accuracy and cross-domain robustness, while avoiding the overhead of explicit hierarchical networks.

Inspired by the hyperbolic image segmentation [44] method, this article introduces hyperbolic geometric structures into remote sensing semantic-segmentation tasks and conducts pixelwise classification modeling. Unlike traditional methods that perform pixel-level category prediction in Euclidean space, this article projects the extracted image features $\mathbf{F}$ into the negatively curved Poincaré Ball space $\mathbb{D}_c^n$ via the exponential mapping $\mathrm{Exp}_0^c(\cdot)$, thereby obtaining an embedded representation with hyperbolic structure constraints.

In hyperbolic space, to achieve effective category discrimination, we use gyroplanes as discriminant hypersurfaces between categories. For each category, the geometric definition of gyroplanes constructs corresponding classification boundaries in hyperbolic manifolds. Specifically, the model calculates the logit based on the hyperbolic distance between embedded points and gyroplanes of various categories, and obtains the final pixel-level classification probability through softmax. During training, we use the standard cross-entropy loss function to supervise the logit, with the optimization objective being to maximize the conditional log-likelihood. Finally, we calculate the network's loss using the cross-entropy loss function

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W}\log p_{ij,y_{ij}} \tag{22}$$

where H and W are the height and width of the input image, respectively, and $p_{ij,y_{ij}}$ denotes the model's prediction probability of the true category y_{ij} at pixel (i, j). By maximizing the prediction probability of the true category, the cross-entropy loss can effectively guide the model to learn the semantic classification ability at the pixel level and optimize the overall segmentation performance.

It is worth noting that this article is the first work in the field of remote sensing semantic segmentation to apply hyperbolic geometry to flattened category learning without the need to construct hierarchical category labels. By introducing a modeling framework based on hyperbolic space, we not only explicitly encode the geometric structural differences between classes at the representation level but also leverage the concavity feature of hyperbolic distance compared to Euclidean distance. This enables more effective distinction between categories with semantic hierarchies or spatial relationships. This characteristic endows hyperbolic embeddings with stronger structural awareness and generalization potential when modeling interclass relationships in remote sensing scenarios.

IV. Experiments

A. Experimental Setup

1) Model Train and Inference: Our unified multitask learning framework is developed on Ubuntu 20.04.1 LTS (64 bit), implemented with Python 3.8, PyTorch 2.0, and CUDA 11.8. The experimental hardware environment consists of two NVIDIA RTX 4090 GPUs, Intel[1] Xeon[1] Gold 6330 CPU at 2.00-GHz processor, and 128-GB system memory. To ensure experimental reproducibility, we provide detailed model configurations: all hyperparameters are set according to the original articles of the corresponding methods. During training, we use the Adam optimizer with a learning rate of 0.001 and a batch size of 2, running for 160 000 iters.

B. Datasets

To comprehensively evaluate the performance of our proposed remote sensing semantic-segmentation method, we conducted experiments on four widely used remote sensing datasets: iSAID [52], LoveDA [53], Potsdam, and Vaihingen [54]. These datasets cover different spatial resolutions, scene complexities, and land-cover types.

Through experiments on these datasets, we are able to comprehensively validate the effectiveness and adaptability of the proposed method in different remote sensing scenarios. The experimental results will demonstrate how our method significantly improves the performance of the model in remote sensing semantic-segmentation tasks while reducing the training parameters.

C. Evaluation Metrics

To comprehensively evaluate the performance of semantic-segmentation models, this article uses three commonly used evaluation metrics: average pixel accuracy (aAcc), average IoU (mIoU), and average $F1$-score (mFscore). The specific definitions are as follows.

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TABLE I
COMPARISONS OF DIFFERENT VFM BACKBONES ON FOUR REMOTE SENSING BENCHMARKS

Methods	Backbone	Trainable Params(M)	mIoU (%) iSAID	LoveDA	Potsdam	Vaihingen
FCN(2016) [20]	R50	47.12	56.75	50.09	76.07	91.33
DeepLabV3(2017) [23]	R50	65.74	53.83	51.37	78.83	95.02
DeepLabV3+(2018) [24]	R50	41.21	51.54	51.59	78.87	94.91
SegFormer(2021) [35]	MIT-B5	81.97	65.16	52.53	79.02	94.01
UperNet(2018) [36]	R50	64.04	55.48	50.55	77.71	89.88
Swin-UperNet(2023) [25]	Swin-B[a]	120	62.65	53.23	79.60	92.45
SegNeXt(2022) [26]	MSCAN-B	27.56	60.77	52.04	79.23	92.59
SETR(2021) [34]	ViT-L[b]	311	56.87	53.33	75.99	83.56
Mask2Former(2022) [37]	R50	44	58.82	52.54	75.17	86.16
RS3mamba(2024) [57]	R18-Mamba-T	42.81	30.13	50.93	74.40	82.78
SACANet(2023) [58]	HRNetv2 [60]	30.31	52.64	49.41	78.71	89.60
LOGCAN++(2025) [59]	MobileNetV3-L	25.23	45.91	51.64	79.45	93.40
HyperR3SNet(Ours)	ViT-L[b]$(DinoV2)$	**16.65**	67.28	55.13	79.90	95.92
HyperR3SNet(Ours)	ViT-L[b]$(DinoV3)$	**16.65**	**67.60**	**55.86**	**80.07**	**96.46**

[a] Swin-B: Swin Transformer Base
[b] ViT-L: Vision Transformer Large

1) *Average Pixel Accuracy (aAcc):*

$$\text{aAcc} = \frac{\sum_{i=1}^{C} TP_i}{\sum_{i=1}^{C} (TP_i + FP_i + FN_i)} \tag{23}$$

where C represents the total number of categories, TP_i denotes the number of correctly predicted pixels for category i, and FP_i and FN_i represent the number of incorrectly predicted pixels for category i, respectively. aAcc measures the overall prediction accuracy of the model at the pixel level across all categories.

2) *mIoU:*

$$\text{mIoU} = \frac{1}{C} \sum_{i=1}^{C} \frac{TP_i}{TP_i + FP_i + FN_i}. \tag{24}$$

mIoU measures the IoU ratio between the predicted region and the true region for each category. A higher value indicates a higher degree of overlap between the predicted results and the true labels, making it one of the core metrics for evaluating the performance of segmentation models.

3) *Mean F1-Score (mFscore):*

$$\text{mFscore} = \frac{1}{C} \sum_{i=1}^{C} \frac{2 \cdot TP_i}{2 \cdot TP_i + FP_i + FN_i}. \tag{25}$$

mFscore measures the balance between precision and recall for each category, comprehensively reflecting the overall performance of the model in category prediction.

D. Comparison of VFM Capabilities

As shown in Table II, under a comparable parameter budget (CLIP, EVA02, DINOv2, and DINOv3 all adopt ViT-Large backbones with roughly 3×10^8 parameters), the DINO family—especially DINOv3-Large—consistently achieves the best overall performance across the four remote sensing benchmarks. Concretely, DINOv3-Large attains mIoU scores of 51.98, 51.88, 73.69, and 78.99 on iSAID, LoveDA, Potsdam, and Vaihingen, respectively, outperforming CLIP (ViT-Large) and EVA02 (Large), and even surpassing SAM (Huge), which has a substantially larger parameter count.

TABLE II
COMPARISON WITH THE SOTA SEMANTIC-SEGMENTATION METHODS

Method	Backbone	Params (M)	iSAID	LoveDA	Potsdam	Vaihingen
CLIP	ViT-Large	316.74	51.52	50.88	71.09	77.13
EVA02	ViT-Large	304.24	51.92	49.58	71.48	74.57
SAM	ViT-Huge	632.18	46.21	50.36	71.93	77.07
DINOv2	ViT-Large	**304.20**	50.78	50.38	72.53	77.01
DINOv3	ViT-Large	**304.20**	**51.98**	**51.88**	**73.69**	**78.99**

These results suggest that, compared with pretraining paradigms that are more tailored to vision–language alignment or interactive segmentation (e.g., CLIP and SAM), the self-supervised DINO series learns more transferable mid- and high-level semantic and structural information that is particularly suitable for dense prediction tasks. Given the same downstream decoder and training protocol, DINOv2/DINOv3, especially DINOv3, provide more discriminative and structurally coherent feature representations, leading to higher segmentation accuracy and more stable cross-dataset generalization in diverse remote sensing scenarios. Therefore, we adopt DINOv3-Large as the default vision foundation backbone in all subsequent experiments to fully exploit its strong representation capacity.

E. Comparison With SOTA Methods

To better evaluate the capabilities of our proposed Hyper3RSNet method in remote sensing interpretation, we conducted comprehensive experiments in remote sensing interpretation scenarios. The input image size was uniformly set to 512×512, and we compared our method with previous SOTA methods across multiple datasets, demonstrating the generalizability and robustness of our method. As shown in Table II, our method outperforms previous SOTA methods

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TABLE III

COMPARISON OF SEMANTIC-SEGMENTATION METHODS ON iSAID DATASET

Methods	Backbone	aAcc (%)	mIoU (%)	mFscore (%)	Trainable Params(M)
FCN [20]	R50	98.65	56.75	70.39	47.13
DeepLabV3 [23]	R50	98.62	53.83	66.73	65.74
DeepLabV3+ [24]	R50	98.50	51.54	64.28	41.22
Segformer [35]	MIT-B5	98.98	65.16	77.61	81.97
UPerNet [36]	R50	98.77	55.48	72.48	64.04
Swin-UPerNet [25]	Swin-B	98.91	62.65	75.49	120
SegNeXt [26]	MSCAN-B	98.86	60.77	73.91	27.56
SETR [34]	ViT-L	98.69	56.87	70.24	311
Mask2Former [37]	R50	98.82	58.82	71.85	44
RS3mamba [57]	R18-Mamba-T	97.70	30.13	57.10	42.81
SACANet [58]	HRNetv2-w32 [61]	98.72	52.64	65.82	30.31
LOGCAN++ [59]	MobileNetV3-L	98.54	45.91	58.92	25.23
HyperR3SNet (Ours)	ViT-L($DinoV2$)	99.02	67.28	79.22	**16.65**
HyperR3SNet (Ours)	ViT-L($DinoV3$)	**99.03**	**67.60**	**79.54**	**16.65**

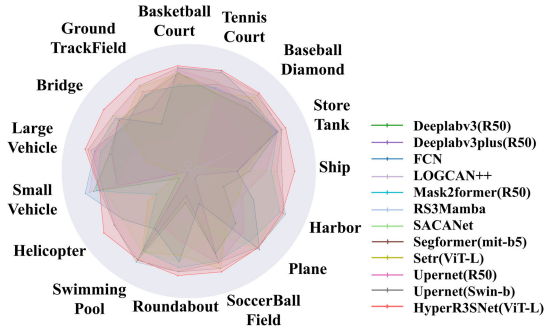


Fig. 3. Comparison of visualization effects on iSAID datasets. Our method achieved the best recognition results for different categories on all iSAID datasets. (a) Image. (b) Ground truth. (c) SegFormer. (d) UPerNet. (e) Mask2Former. (f) SETR. (g) HyperR3SNet (Ours).

in terms of mIoU metrics across four datasets. Additionally, among the compared methods, our model has the fewest training parameters at 16.65M. This fully demonstrates the excellent generalization and robustness of our method.

1) Quantitative Analysis on the iSAID: In the evaluation experiments on the iSAID dataset, as shown in Table III, we systematically compared the performance of the proposed HyperR3SNet method with several mainstream semantic-segmentation models on the iSAID dataset. HyperR3SNet achieved the best performance across all three evaluation metrics: overall pixel accuracy (aAcc) reached 99.03%, mIoU was 67.60%, and $F1$-score (mFscore) was 79.54%. The mIoU metric outperforms the second-best model by 2.44 percentage points. The visualization results of the inference are shown in Fig. 3.

As shown in Fig. 4, we can also see that in the category-level $F1$-score comparison conducted on the iSAID dataset, our proposed HyperR3SNet method significantly outperforms existing mainstream methods in the vast majority of categories, demonstrating its excellent semantic recognition capabilities and category generalization capabilities. Specifically, HyperR3SNet achieves the best $F1$-scores in target categories such as "Basketball Court," "Tennis Court," "Baseball Diamond," "Bridge," and "Large Vehicle." In the "Bridge" category, it outperforms the second-best method by 8.24 percentage points, maintaining a high level of performance across all categories.

In contrast, other methods exhibit significant performance fluctuations across multiple categories, particularly showing instability in small and edge targets (e.g., "Helicopter" and "Roundabout"). HyperR3SNet, however, leverages its structure-aware mechanism and hyperbolic space modeling capabilities to effectively enhance semantic consistency and spatial recognition capabilities, demonstrating stable and comprehensive recognition advantages.

TABLE IV

OVERALL PERFORMANCE COMPARISON ON THE LOVEDA DATASET

Methods	Backbone	aAcc (%)	mIoU (%)	mFscore (%)	Trainable Params(M)
FCN [20]	R50	67.94	50.09	65.88	47.13
DeepLabV3 [23]	R50	69.86	51.37	66.83	65.74
DeepLabV3+ [24]	R50	69.72	51.59	67.16	41.22
Segformer [35]	MIT-B5	69.97	52.53	67.79	81.97
UPerNet [36]	R50	68.80	50.55	66.16	64.04
Swin-UPerNet [25]	Swin-B	70.31	53.23	68.42	120
SegNeXt [26]	MSCAN-B	70.22	52.04	67.42	27.56
SETR [34]	ViT-L	70.41	53.33	65.36	311
Mask2Former [37]	R50	70.73	52.54	67.91	44
RS3mamba [57]	R18-Mamba-T	67.70	50.93	66.86	42.81
SACANet [58]	HRNetv2-w32 [61]	69.63	49.41	64.46	30.31
LOGCAN++ [59]	MobileNetV3-L	70.31	51.64	66.67	25.23
HyperR3SNet(Ours)	ViT-L($DinoV2$)	71.71	55.13	70.11	**16.65**
HyperR3SNet(Ours)	ViT-L($DinoV3$)	**72.34**	**55.86**	**70.82**	**16.65**

TABLE V

PER-CLASS IOU COMPARISON ON THE LOVEDA DATASET

Methods	IoU (%)					
	Building	Road	Water	Barren	Forest	Agricultural
FCN [20]	63.11	53.58	64.27	27.70	41.94	47.53
DeepLabV3 [23]	64.87	55.50	66.38	27.36	38.65	52.49
DeepLabV3+ [24]	65.15	55.19	65.59	29.01	40.53	51.07
Segformer [35]	66.85	56.30	70.45	28.66	39.59	53.51
UPerNet [36]	63.43	54.25	66.69	27.34	39.40	49.22
Swin-UPerNet [25]	66.30	57.22	**72.57**	28.83	42.81	51.33
SegNeXt [26]	64.52	58.00	67.45	27.14	40.94	51.91
SETR [34]	65.39	56.46	71.80	27.85	44.67	56.18
Mask2Former [37]	64.54	55.72	69.87	30.06	39.20	53.69
RS3mamba [57]	58.75	57.92	61.00	**37.24**	39.67	33.98
SACANet [58]	59.62	54.86	70.23	21.63	35.84	54.27
LOGCAN++ [59]	65.55	55.21	70.19	25.41	39.01	54.5
HyperR3SNet($DinoV2$)	68.25	59.02	71.90	30.36	46.15	58.33
HyperR3SNet($DinoV3$)	**69.02**	**59.76**	71.49	33.52	44.79	**60.10**

2) Quantitative Analysis on the LoveDA: In the evaluation experiments on the LoveDA dataset, as shown in Tables IV and V, we systematically compared the proposed HyperR3SNet method with several mainstream semantic-segmentation methods on the LoveDA dataset. Under a unified input configuration and training strategy, HyperR3SNet achieved leading performance across all evaluation metrics, demonstrating excellent perception capabilities in urban–rural mixed scenarios. Specifically, HyperR3SNet achieved an overall accuracy (aAcc) of 71.71%, an mIoU of 55.13%, and an $F1$-score of 70.11%, all of which significantly outperform existing methods. Compared to the closest competing methods, UPerNet (Swin-B, mIoU of 53.23%) and SETR (mIoU of 53.33%), our method improves mIoU by 1.9 and 1.8 percentage points, respectively, and achieves at least a 1.7 advantage in $F1$-score. The visualization results of the inference are shown in Fig. 5.

To comprehensively evaluate the model's segmentation capabilities across different categories, we conducted a

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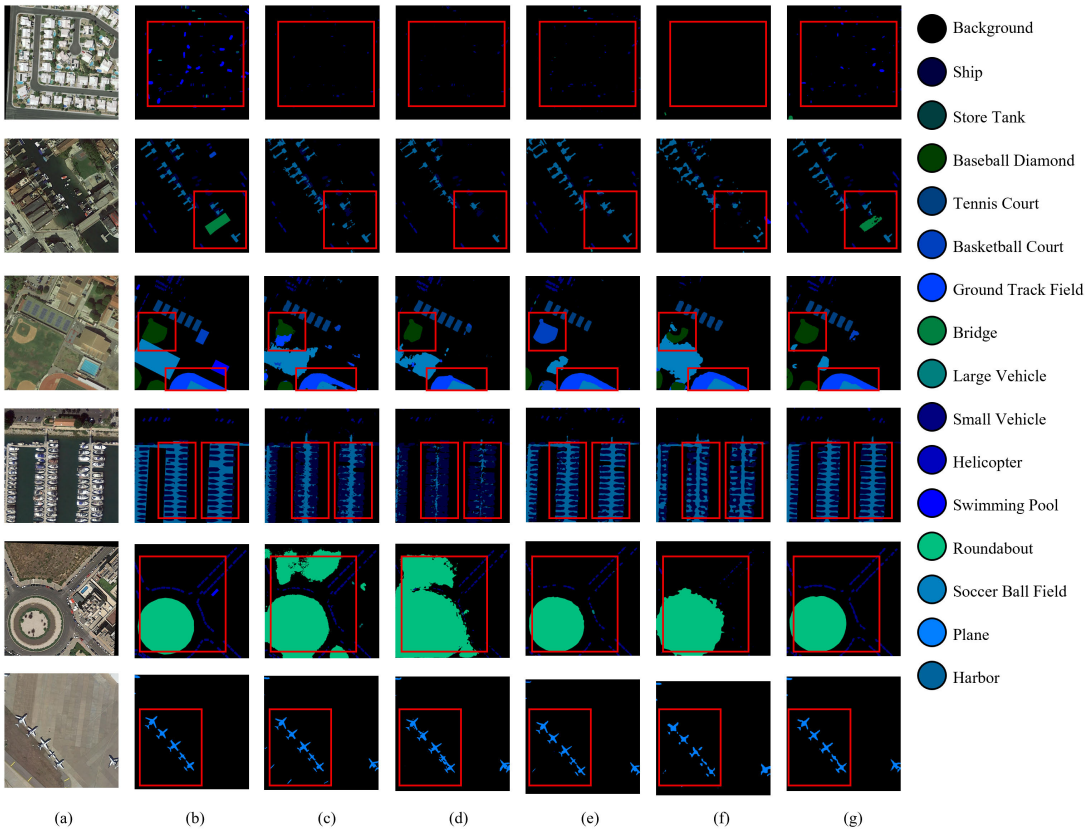


Fig. 4. Categorywise $F1$-score distribution on iSAID.

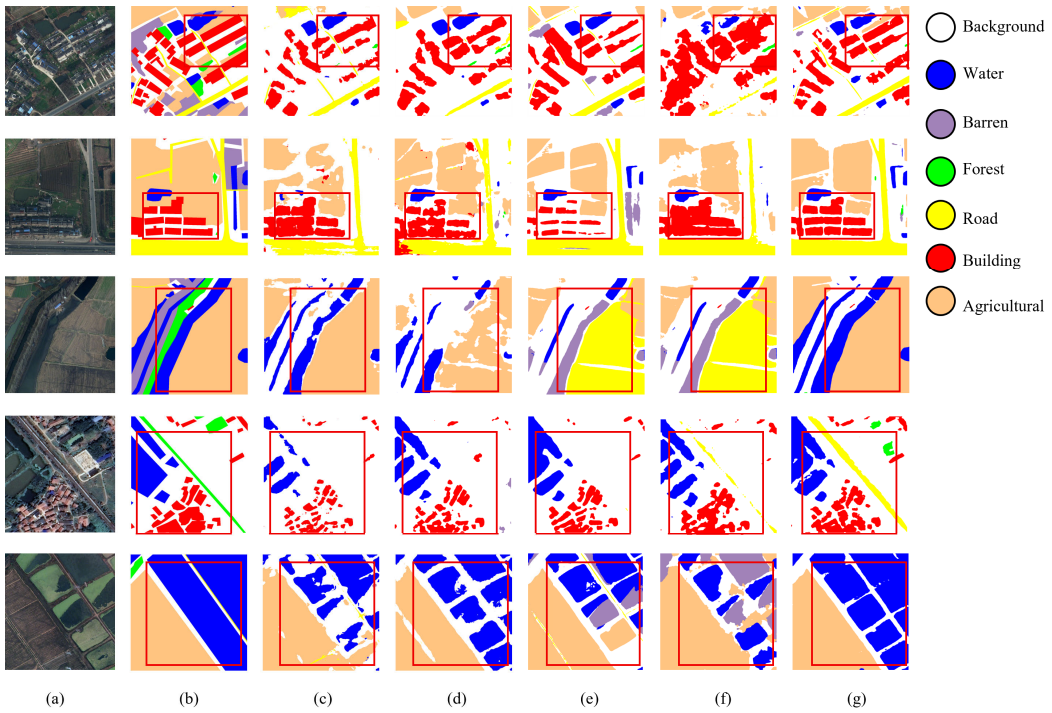


Fig. 5. Comparison of visualization effects on LoveDA datasets. Our method achieved the best recognition results for different categories on all LoveDA datasets. (a) Image. (b) Ground truth. (c) SegFormer. (d) UPerNet. (e) Mask2Former. (f) SETR. (g) HyperR3SNet (Ours).

detailed comparative analysis of the IoU metrics for the six key semantic categories in the LoveDA dataset. The results are shown in Tables IV and V. As can be seen, our proposed HyperR3SNet achieved the highest IoU in five out of the

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TABLE VI
COMPARISON OF SEMANTIC-SEGMENTATION METHODS ON POTSDAM AND VAIHINGEN DATASETS

Methods	Backbone	Trainable Params (M)	Potsdam			Vaihingen		
			aAcc (%)	mIoU (%)	mFscore (%)	aAcc (%)	mIoU (%)	mFscore (%)
FCN [20]	R50	47.13	89.94	76.07	86.03	96.80	91.33	95.39
DeepLabV3 [23]	R50	65.74	90.30	78.83	87.82	98.23	95.02	97.42
DeepLabV3+ [24]	R50	41.22	90.24	78.87	87.74	98.15	94.91	97.36
Segformer [35]	MIT-B5	81.97	90.32	79.02	87.98	97.81	94.01	96.88
UPerNet [36]	R50	64.04	89.78	77.71	87.07	96.16	89.88	94.59
Swin-UPerNet [25]	Swin-B	120	90.70	79.60	88.33	97.06	92.45	96.03
SegNeXt [26]	MSCAN-B	27.56	90.43	79.23	88.10	97.15	92.59	96.10
SETR [34]	ViT-L	311	89.11	75.99	85.95	93.44	83.56	90.81
Mask2Former [37]	R50	44	88.00	75.17	85.42	93.73	86.16	92.47
RS3mamba [57]	R18-Mamba-T	42.81	87.82	74.40	84.87	95.63	82.78	90.34
SACANet [58]	HRNetv2-w32 [60]	30.31	88.46	78.71	87.94	95.74	89.60	94.44
LOGCAN++ [59]	MobileNetV3-L	25.23	88.79	79.45	88.40	97.61	93.40	96.54
HyperR3SNet(Ours)	ViT-L($DinoV2$)	**16.65**	**90.77**	79.90	88.54	98.57	95.92	97.90
HyperR3SNet(Ours)	ViT-L($DinoV3$)	**16.65**	**90.77**	**80.07**	**88.67**	**98.74**	**96.46**	**98.19**

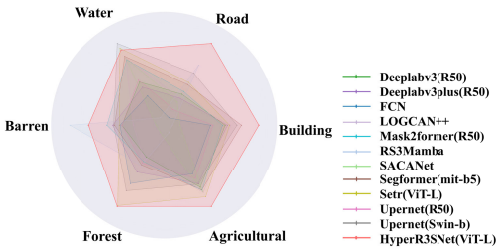


Fig. 6. Categorywise $F1$-score distribution on LoveDA.

six categories, demonstrating significantly superior overall performance compared to existing methods.

Specifically, HyperR3SNet achieved 68.25% in the "Building" category, an improvement of 5.14 percentage points over the traditional baseline FCN and 1.95 percentage points over the current highest Swin-B UPerNet. In the "Road" category, it achieved 59.02%, an improvement of 2.72 percentage points over SegFormer and 1.8 percentage points over Swin-B UPerNet. Additionally, in the "Barren" category, it achieved an IoU of 30.36%, an improvement of over 3 percentage points compared to the 27%–28% range of most methods.

More notably, the IoU for the "Forest" category reached 46.15%, an increase of 6.56 percentage points compared to SegFormer, demonstrating significant semantic modeling capabilities in complex natural scenes. The IoU for the "Agricultural" category is 58.33%, which is 5.84 percentage points higher than DeepLabV3 and 2.15 percentage points higher than the strongest SETR, making it particularly outstanding.

Although our method performs slightly worse than Swin-B UPerNet in the "Water" category (71.90% versus 72.57%), it achieves significant breakthroughs in the remaining five categories, demonstrating broader and more stable overall improvements. As shown in Fig. 6, our method also significantly outperforms previous SOTA methods in terms of $F1$-scores.

In summary, HyperR3SNet demonstrates comprehensive and quantitatively significant performance improvements in the LoveDA multicategory scenario, validating the effectiveness of its structure-aware enhancement and low-dimensional hyperbolic embedding mechanism in urban–rural mixed semantic parsing.

3) Quantitative Analysis on Potsdam and Vaihingen: To evaluate the generalization capability of HyperR3SNet across different remote sensing scenarios, we selected two sets of ISPRS high-resolution data: Potsdam with a resolution of 5 cm/pixel and Vaihingen with a resolution of 9 cm/pixel. The former features diverse urban landscapes and wide roads, while the latter focuses on a German town with densely packed roofs and complex textures. The experimental results are shown in Table VI: in Potsdam, HyperR3SNet achieved an overall accuracy of 90.77% and an mIoU of 79.90%, outperforming Swin-B UPerNet by 0.30 percentage points and SegFormer by 0.82 percentage points. while its parameter count is only 23.24M, significantly lighter than the 120M of Swin-B and 311M of SETR. In Vaihingen, the model also maintains its lead, achieving 98.57% aAcc, 95.92% mIoU, and 97.90% $F1$, with mIoU improving by 0.9 and 1.91 percentage points over DeepLabV3 and SegFormer, respectively. The visualization results of the inference are shown in Fig. 7. Categorywise comparison (Fig. 8) shows: in low-resolution scenes, it leads in all categories; in high-resolution scenes, it ranks first in all categories except "impervious surfaces" and "buildings".

Furthermore, on the Vaihingen dataset, HyperR3SNet also achieved the best performance across all metrics: aAcc reached 98.57%, mIoU was 95.92%, and the $F1$-score reached 97.90%, which is 0.9% higher than DeepLabV3 and 1.91% higher than SegFormer. This result clearly demonstrates that HyperR3SNet is not only capable of handling wide-area, structurally complex urban neighborhood images but can also accurately process high-density, texture-rich building distribution scenarios, showcasing its strong adaptability and boundary analysis capabilities.

In summary, our method maintains leading performance in a variety of remote sensing scenarios while having a low number of training parameters, verifying its potential for dual optimization of spatial resolution accuracy and computing resources in practical applications.

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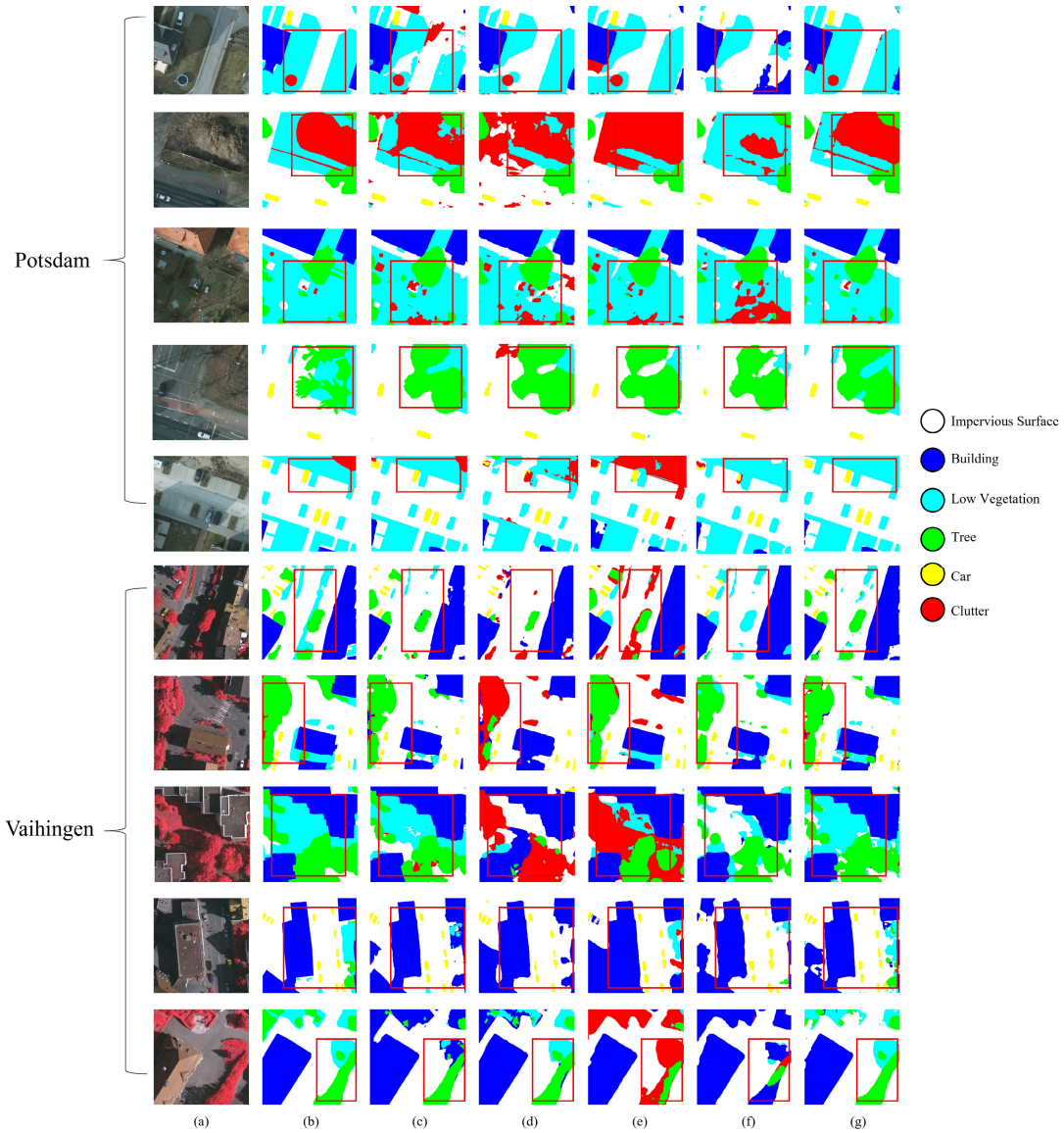


Fig. 7. Comparison of visualization effects on the Potsdam and Vaihingen datasets. Our method achieved the best recognition results for different categories on all Vaihingen datasets. (a) Image. (b) Ground truth. (c) SegFormer. (d) UPerNet. (e) Mask2Former. (f) SETR. (g) HyperR3SNet (Ours).

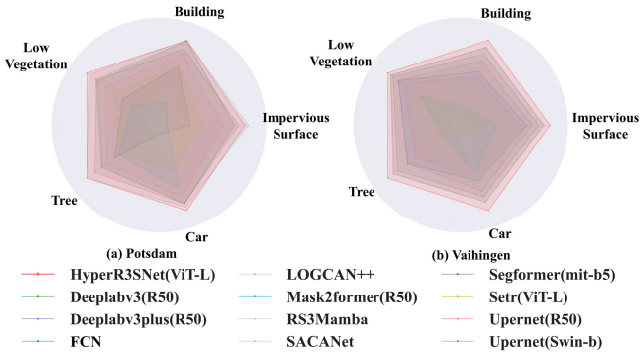


Fig. 8. Categorywise $F1$-score distribution on Potsdam and Vaihingen datasets.

F. Ablation Studies

To validate the effectiveness of the MSCA module in improving object classification accuracy, we conducted ablation experiments on four typical remote sensing datasets (iSAID, LoveDA, Potsdam, and Vaihingen), with the results shown in Table VII. Without the MSCA module, HyperR3SNet achieved mIoU values of 51.98%, 51.88%, 73.69%, and 78.99% on the four datasets, respectively. After incorporating this module, the model achieved significant performance improvements across all datasets, reaching 63.24%, 53.77%, 78.99%, and 93.29%, respectively. Specifically, on the

TABLE VII

ABLATION OF MSCA AND HYPERBOLIC PIXELWISE LOSS ON FOUR BENCHMARKS (WHEN THE HYPERBOLIC MODULE IS DISABLED, STANDARD EUCLIDEAN PIXELWISE CLASSIFICATION IS ADOPTED BY DEFAULT)

Method	MSCA	Hyperbolic	IoU (%)			
			iSAID	LoveDA	Potsdam	Vaihingen
HyperR3SNet (baseline)	–	–	51.98	51.88	73.69	78.99
+ MSCA	✓	–	63.24	53.77	78.99	93.29
+ Hyperbolic	✓	✓	**67.28**	**55.13**	**79.90**	**95.92**

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TABLE VIII
ABLATION STUDY ON THE CURVATURE PARAMETER c OF THE HYPERBOLIC MODULE. THE SECTIONAL CURVATURE OF THE POINCARÉ BALL EQUALS $-c$. THE DEFAULT SETTING USED IN ALL MAIN EXPERIMENTS IS $c = 1$

Curvature c	iSAID	LoveDA	Potsdam	Vaihingen
0.1	67.45	55.12	79.79	96.35
0.3	67.32	55.05	79.97	96.24
1 (default)	67.60	**55.86**	80.07	**96.46**
3	**67.77**	55.27	79.88	96.37
10	67.03	55.52	**80.15**	96.28

iSAID and Vaihingen datasets, the improvements were 11.26% and 14.30%, respectively, indicating that this module has a significant advantage in addressing issues of scale inconsistency and target deformation in remote sensing imagery. It is worth noting that in this setting, the model still adopts standard Euclidean pixelwise classification.

To further validate the effectiveness of the hyperbolic space embedding mechanism in improving the classification performance of remote sensing objects, we conducted ablation experiments on four typical remote sensing datasets (iSAID, LoveDA, Potsdam, and Vaihingen), with the results shown in Table VII. By replacing the default Euclidean pixelwise classification with the proposed hyperbolic classification, while keeping the MSCA module unchanged, it can be observed that the introduction of this module significantly improves classification performance across all datasets.

Specifically, on the iSAID dataset, mIoU improved from 63.24% to 67.28%, an increase of 4.04 percentage points, and on the LoveDA dataset, it improved by 1.36 percentage points (from 53.77% to 55.13%). The Potsdam dataset also achieved an improvement from 78.99% to 79.90%, with particularly significant improvements on the Vaihingen dataset, where mIoU increased from 93.29% to 95.92%, representing an increase of 2.63 percentage points. This demonstrates that the hyperbolic space has a stronger enhancement effect on complex structural categories and high-precision remote sensing scenarios.

Influence of Curvature Parameter: The expressive capacity of hyperbolic embedding is inherently governed by the curvature parameter c, which controls the geometric expansion rate of the Poincaré Ball. Although the hyperbolic head improves segmentation accuracy, it is essential to verify whether the performance gain is robust across different curvature regimes. Therefore, we conduct a systematic ablation study over multiple curvature settings to assess the stability of the proposed hyperbolic modeling.

The experimental results in Table VIII show that segmentation performance remains stable across a wide curvature range ($c \in \{0.1, 0.3, 1, 3, 10\}$), with mIoU variations generally within 0.8% on all datasets. This indicates that the performance improvement brought by the hyperbolic module does not rely on delicate curvature tuning, but instead stems from its geometry-aware embedding mechanism. When c is small (e.g., $c = 0.1$), the hyperbolic space approaches Euclidean geometry, slightly weakening hierarchical separation capability. Conversely, when c becomes large (e.g., $c = 10$), the effective radius of the Poincaré Ball shrinks, which may introduce mild boundary saturation effects and marginally affect stability on certain datasets.

Among all tested configurations, $c = 1$ achieves the most balanced and consistent performance across the four benchmarks. Therefore, we adopt $c = 1$ as the default setting in all main experiments, as it provides an effective tradeoff between hierarchical expressiveness and numerical stability.

Beyond the quantitative improvements, the ablation results reflect a clear complementary design philosophy. The MSCA module operates in the Euclidean spatial domain, focusing on *feature enrichment*. It aggregates multiscale contexts to resolve scale variations and target deformations, ensuring that the encoder captures comprehensive visual patterns. In contrast, the hyperbolic module operates in the embedding geometry domain, focusing on *feature regularization*. It projects these rich features into a hyperbolic manifold to better model the implicit hierarchical structure of semantic categories. Therefore, the two components work in tandem: MSCA ensures the *representational power* of the features (handling "what the object looks like"), while the hyperbolic head ensures the *discriminative power* of the embedding space (handling "how the object is classified"). This synergy leads to the optimal performance observed in the joint model.

V. CONCLUSION

This article addresses the issues of limited annotated data and significant domain differences between remote sensing and natural images in remote sensing semantic-segmentation tasks by proposing an efficient new segmentation framework, *HyperR3SNet*. This method fully leverages the powerful representation capabilities of visual foundation models in the natural-image domain, introduces an MSCA mechanism to enhance the modeling capability of scale-varying structures in remote sensing images, and combines a lightweight task head based on matrix decomposition to effectively reduce model training parameters and computational overhead. Additionally, this article innovatively proposes a pixel-level hyperbolic space loss design, which can still model the potential semantic relationships between categories under conditions of flat label structures and insufficient supervision information, thereby improving structural perception and discrimination capabilities. The experimental results on four remote sensing segmentation benchmark datasets—iSAID, LoveDA, Potsdam, and Vaihingen—show that HyperR3SNet achieves the best performance in terms of mIoU, comprehensively surpassing existing SOTA methods and demonstrating superior accuracy and cross-scenario generalization capabilities. To the best of authors' knowledge, this method is the first to introduce hyperbolic space into remote sensing semantic segmentation for pixel-level supervision, providing a new structural modeling approach for remote sensing semantic understanding.

Despite significant performance improvements, this study has the following limitations: dependence on the quality of pretrained weights—the framework assumes that natural-image self-supervised models have sufficient transferability; if the quality of the base weights is insufficient, the

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overall performance may decline; computational resource consumption—although the matrix decomposition head significantly reduces the number of trainable parameters, the overall backbone is still based on a ViT-L-level model, and the inference stage has certain requirements for GPU memory and computing power, which is not conducive to edge deployment; optical remote sensing data-centric—experiments have only been validated on optical RGB aerial/satellite images and have not yet covered multimodal remote sensing scenarios such as SAR, multispectral, or hyperspectral data, leaving the cross-modal generalization capabilities to be further evaluated and expanded. In the future, we will continue to explore the potential of combining visual foundation models with hyperbolic spaces in the field of multimodal remote sensing imagery.

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